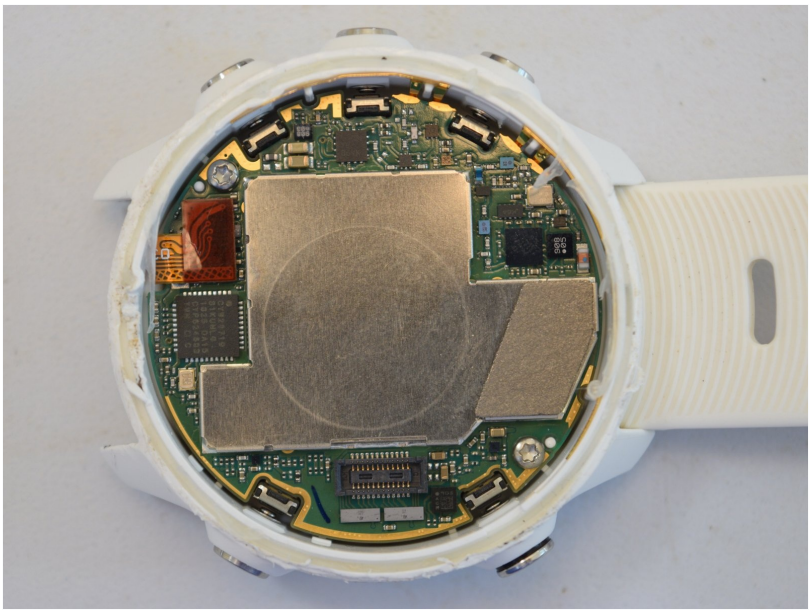
Garmin Forerunner 245 Music Teardown

This watch, like most other recent watches from...

Written By: Kyle Westhaus



INTRODUCTION

This watch, like most other recent watches from Garmin (except for the Fenix series), is glued shut tightly for waterproofing. As such the disassembly is difficult, but not impossible.

TOOLS:

iOpener (1)
Suction Handle (1)
iFixit Opening Tool (1)
Precision Utility Knife (1)
Spudger (1)
T5 Torx Screwdriver (1)

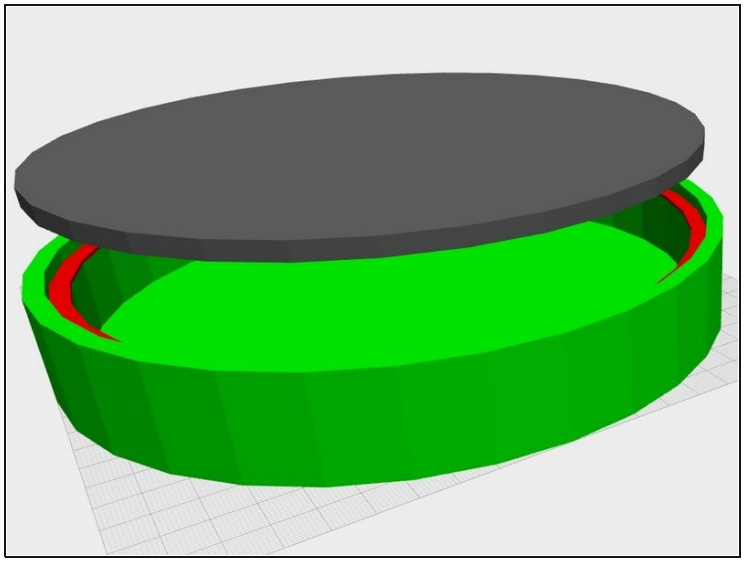
Step 1 — (Optional) Remove bands

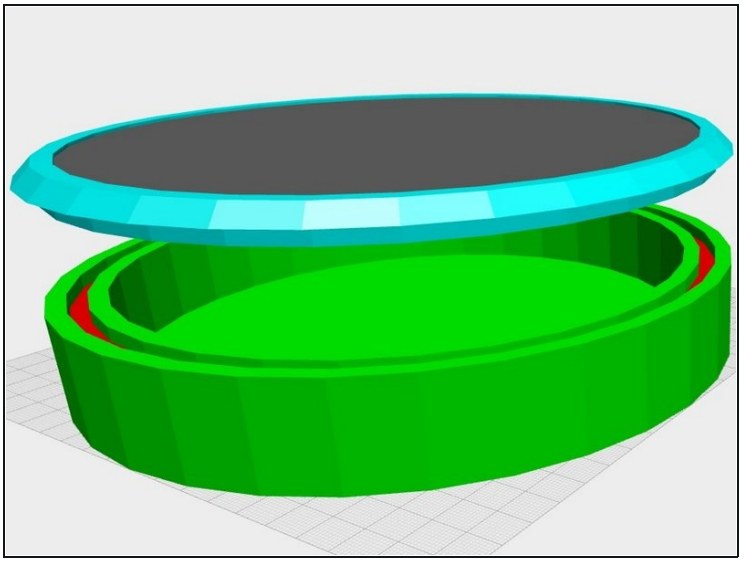




- This watch makes use of 20mm quick release bands.
- Slide the pin on the metal bar and pull the band away from the watch body.

Step 2 — Explanation of glue obstacle





ⓘ (First photo) Past Forerunner models made use of a simplified glue construction which made it easier to heat-soften the glue and remove the screen by prying or by suction.

ⓘ (Second photo) This watch (and other recent models) makes use of a complicated glue construction which makes it more difficult to heat-soften the glue and remove the screen by prying or by suction.

- For more detail, the 3D models used to produce these images are available at these links: (old, new) and can be opened in your browser at https://openjscad.azurewebsites.net/.
  - Watch body
  - Glue
  - Bezel
  - Screen

Step 3 — Attempt 1: Remove the combined screen and bezel via suction





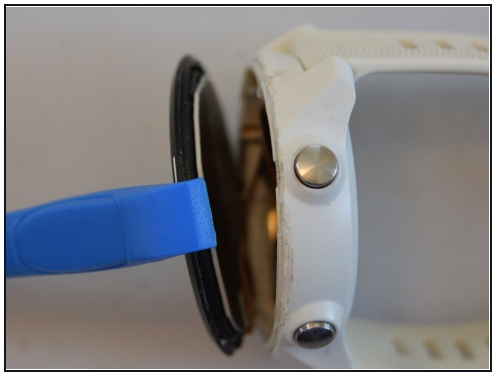
⚠ This method is likely to fail. If you cannot complete it successfully, move on to the (more destructive) next step.

- Utilize your choice of iOpener, heat gun, hair dryer, oven, etc. to soften the internal glue.
- Attach a suction cup to the watch face and pull up while holding the watch body in place until the two separate.

  ⚠ Be careful not to damage the FFC (flexible flat cable) connecting to the screen after separation.

Step 4 — Attempt 2: Remove the combined screen and bezel via prying



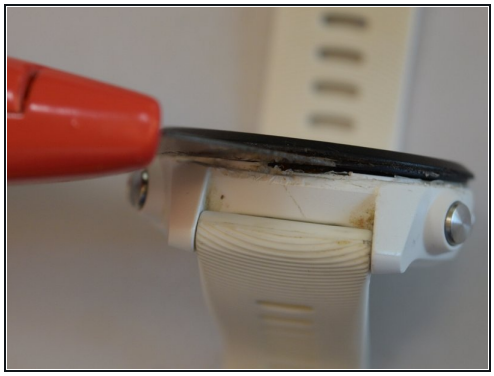
⚠ This method is likely to fail. If you cannot complete it successfully, move on to the (more destructive) next step.

- Utilize your choice of iOpener, heat gun, hair dryer, oven, etc. to soften the internal glue.
- Wedge your tool of choice (spudger, opening tool, knife, etc.) between the bezel brim and the rest of the watch body until you reach the wall of the plastic ring which hangs down from the bezel. Pry the watch face away from the watch body until the glue releases.

⚠ Be careful not to damage the FFC (flexible flat cable) connecting to the screen after separation.

ⓘ You may need to cut some material away from the watch body or bezel brim in order to slide your tool in far enough to pry.

Step 5 — Attempt 3: Remove the combined screen and bezel via cutting





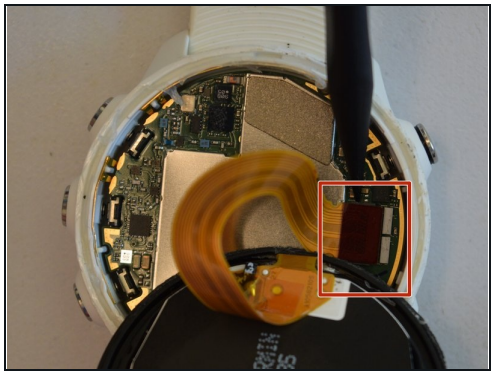
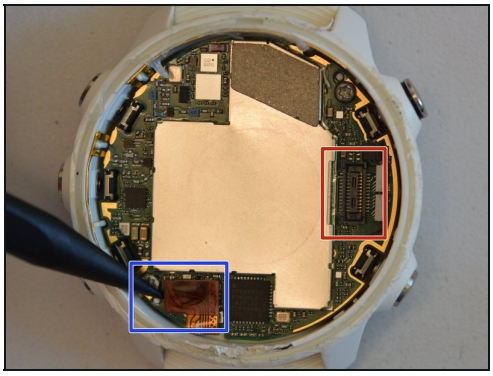
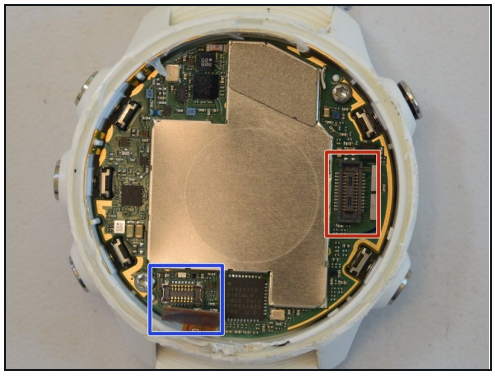
⚠ While this method is more likely to work, it is also more likely to damage the screen.

- Slip a knife between the bezel brim and the rest of the watch body. When you reach the wall of the plastic ring which hangs down from the bezel, don't stop. Instead, saw your way through this wall and make a slot large enough for a pry tool.

  ⚠ The FFC for the screen is particularly vulnerable if you choose to make a slot near the 6 o'clock position (under the "GARMIN" logo) due to its attachment point to the screen just behind the bezel ring.

- Place your prying tool as far as it will go into the new slot and push the screen up until the glue releases.

Step 6 — Disconnect the screen and heart rate sensor

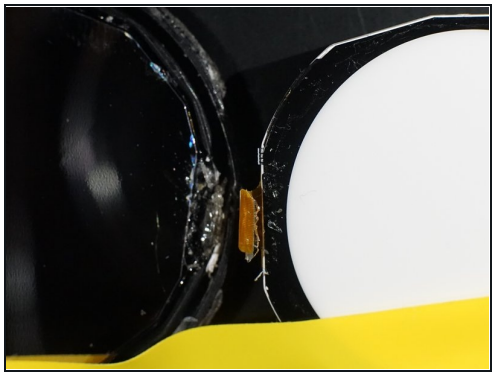




- Disconnect the screen FFC from the motherboard as described in this guide.
- Similarly undo the connector for the FFC leading under the motherboard to the heartbeat sensor and vibration motor.

Step 7 — Screen closeups

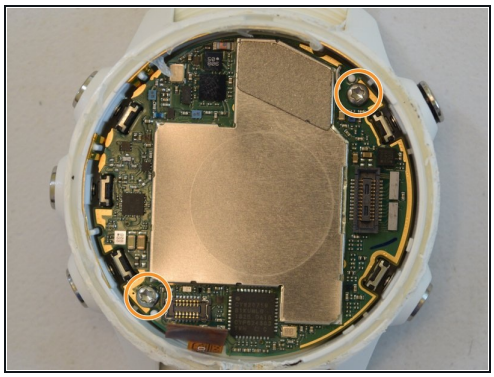


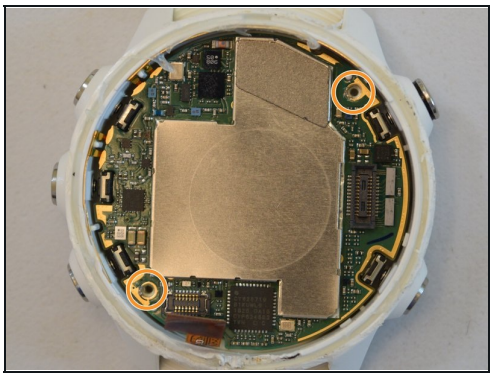


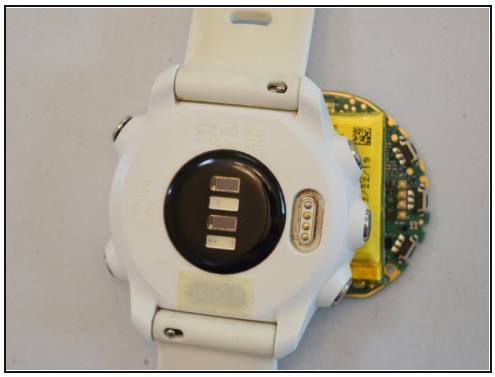


- Insert wisdom here.

Step 8 — Remove the motherboard

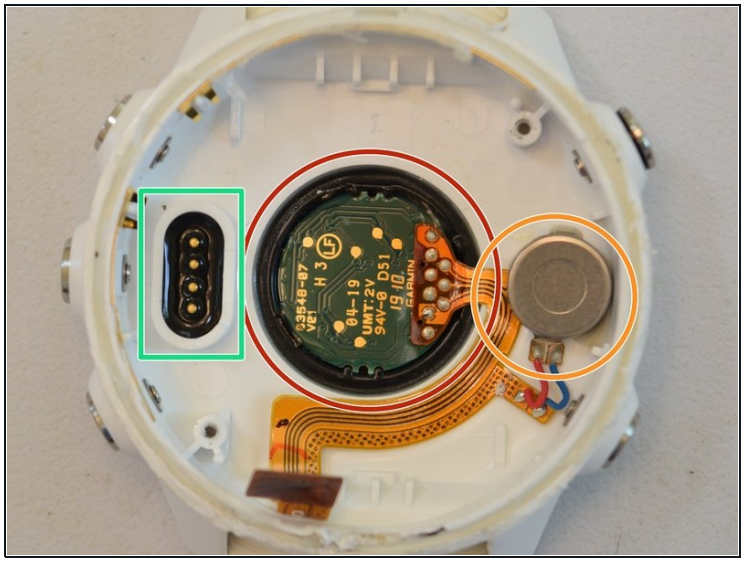


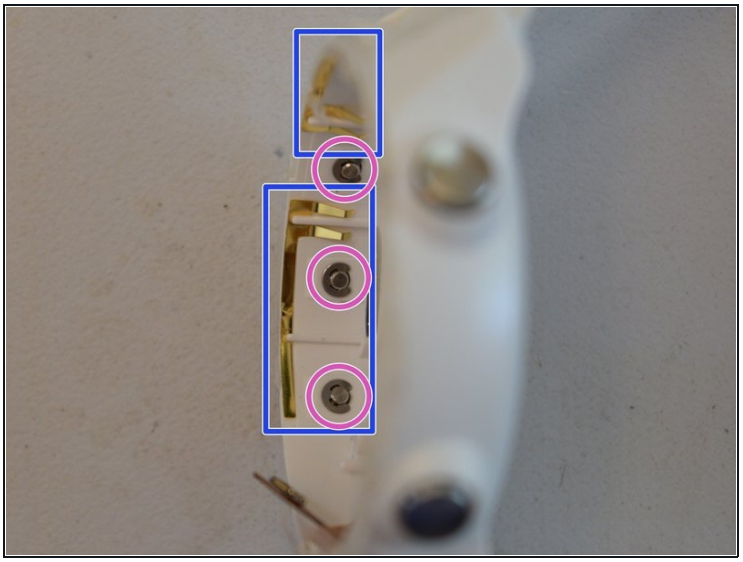




- Undo and remove both of the T5 screws.
- Turn the watch over so it is face down and let the motherboard fall out of the watch body.

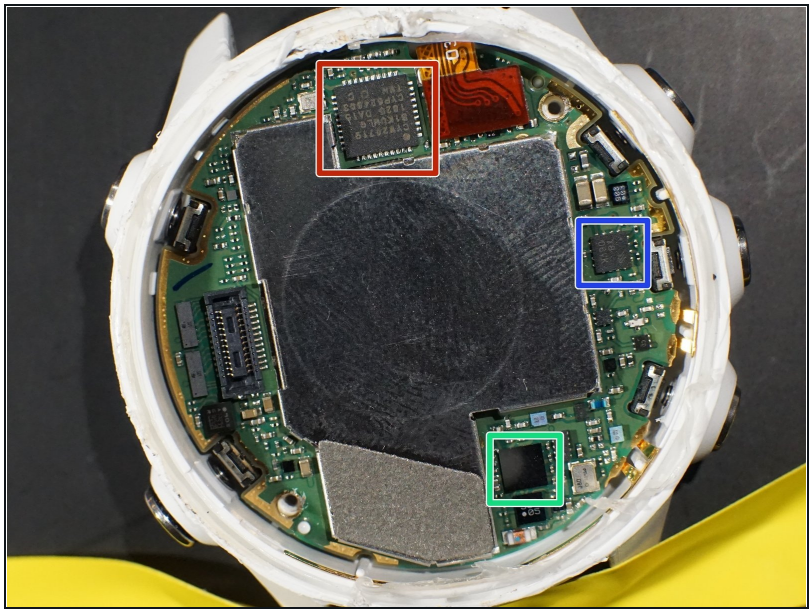
Step 9 — Parts identification (watch body)





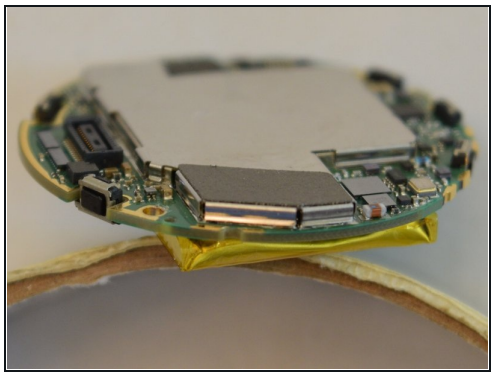
- Heart rate sensor
- Vibration motor
- Charging/USB contacts
- Antennas
- Waterproof button chambers

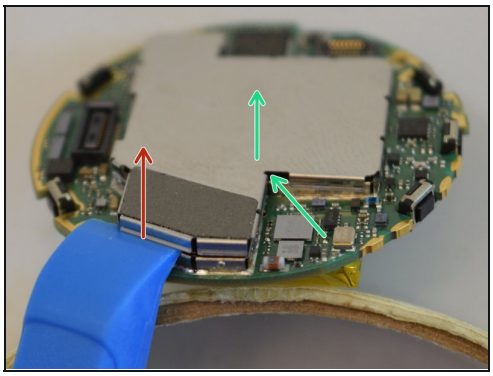
Step 10 — Component identification (outside of shielding)

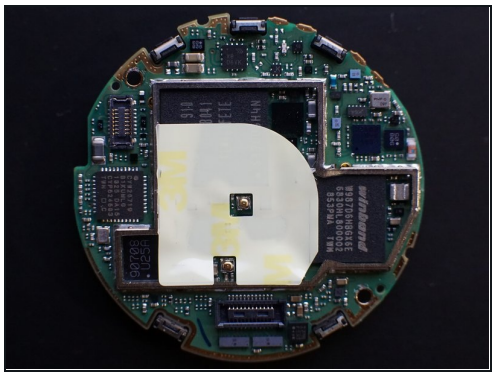


- ANT/BLE/BT coprocessor: Cypress CYW20719 datasheet
- GPS coprocessor: Sony D5603 datasheet
- RF front end: Qorvo (RF Micro Devices) RFFM6205 datasheet

Step 11 — Remove shielding

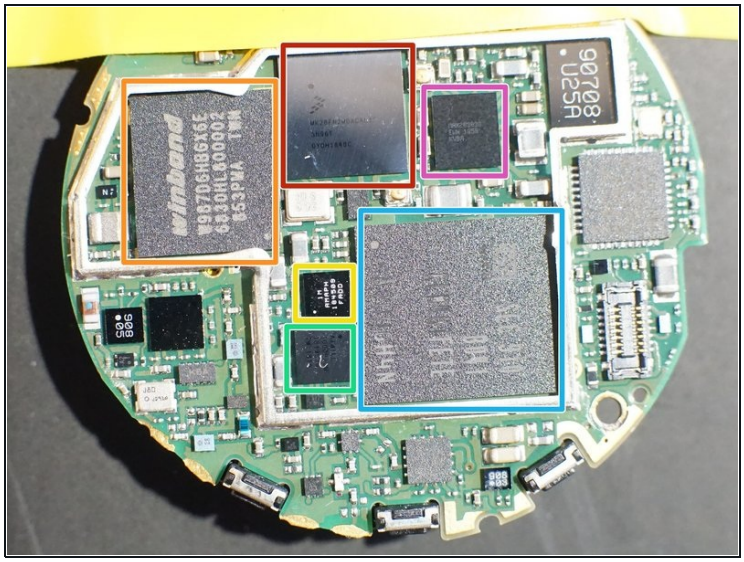


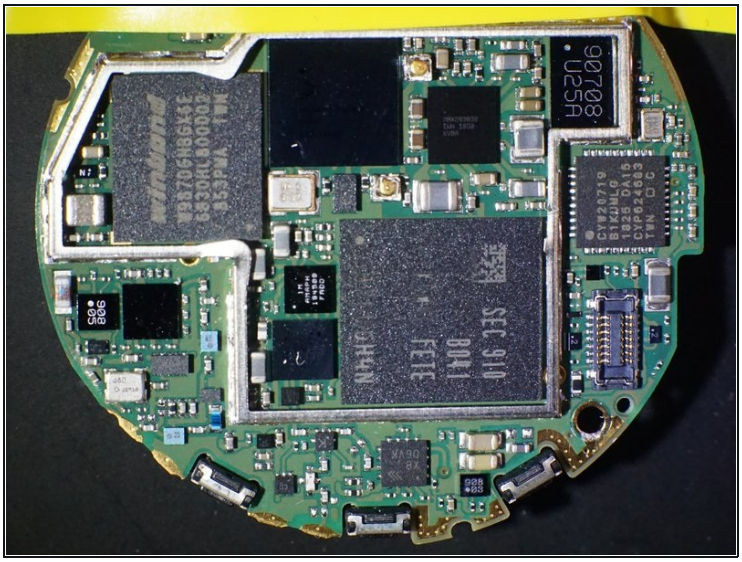




- The sides of the metal shielding are clips which fit around the small metal walls underneath of them on the PCB. You can reach under the end of the clips and push up to remove them.
  - Alternatively, stick something small past the gap between clips and under the shielding, then use it to pull the shielding up.
- Remove the 3M tape covering the rest of the components.

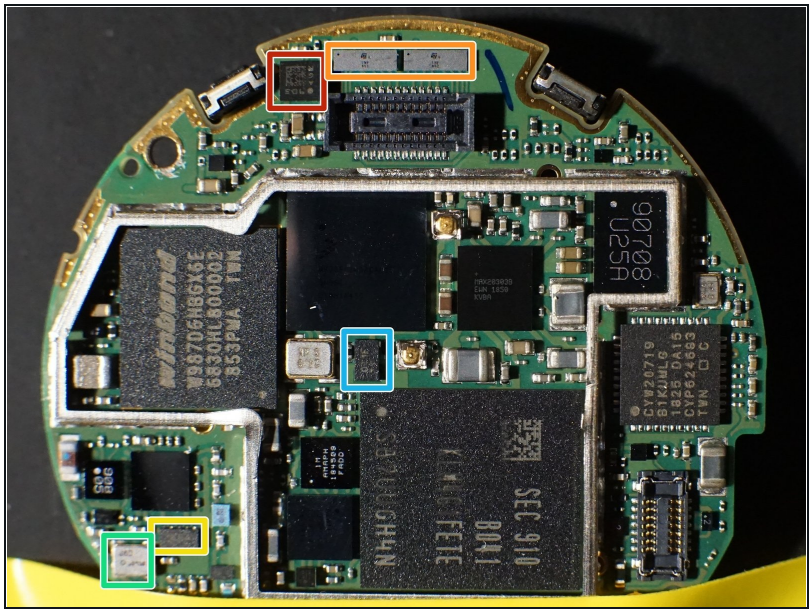
Step 12 — Component identification (inside of shielding)





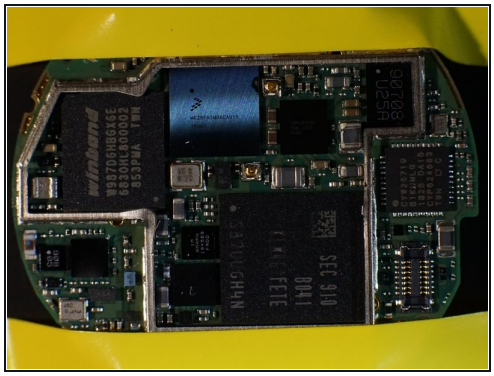
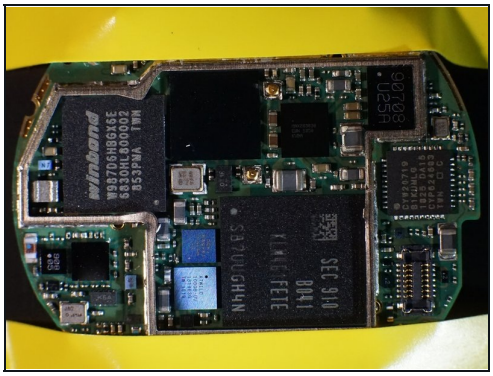
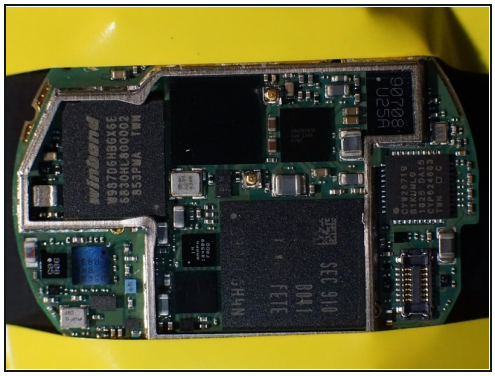
- Main processor: NXP Kinetis MK28FN2M0ACAU15 datasheet, technical reference manual (requires free login)
- SDRAM: Winbond W987D6HBGX6E datasheet
- Sensorhub coprocessor: Ambiq Micro Apollo2 datasheet
- Wifi coprocessor: ATWILC 1000B-UU datasheet
- eMMC: KLM4G1FETE-B041 datasheet
- PMIC: Maxim Integrated MAX20303B datasheet

Step 13 — Remaining components to be identified



- eCompass (accelerometer+magnetometer): STMicro LSM303AGR datasheet thanks Shen!
- LNF 852
- K6A
- J80 (circle) JD910
- Unknown
- Many other chips on both sides of the motherboard are too small to be legible by me, but might make sense to others.
- The chips starting with 908 are likely designed by Garmin and proprietary.
- Other chips with single and double letter names may be named according to this reference.

Step 14 — Further closeups

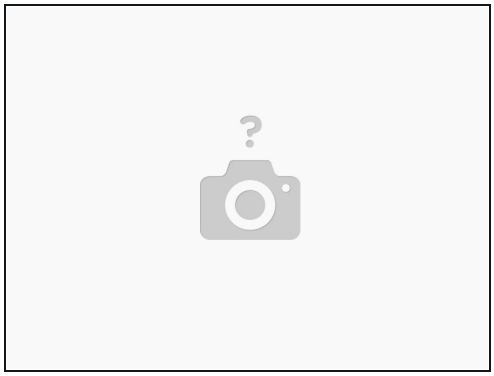
- Insert wisdom here.

Step 15 — Battery



- Battery: Garmin 361-00086-11, 3.8Vcd * 180mAh = 0.68Wh, soldered to motherboard

Step 16 — Going backwards



- Assuming you didn't destroy the watch too much during the teardown, putting it back together should be as simple as following the steps in reverse plus replacing the old glue with some new glue (B-7000 seems to be the most common recommendation online).